BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Inputs, Processes and Outputs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is algorithmic thinking?

2. What does an encryption utility produce?

A) A smaller file
B) A sorted folder
C) Ciphertext
D) Extra storage space

3. What does a compression utility do?

4. A CPU reads an instruction at memory address 120. Correct both claims. Explain why. (a) MAR holds a calculation result. (b) MDR always holds the next instruction's address.

Today's objectives

Identify inputs, processing and output for a specified problem. Construct an accurate IPO diagram.

Key vocabulary - Read, cover, recall

Input
Data supplied to a program.

Processing
Operations or decisions performed on data.

Output
Result produced by an algorithm.

IPO
Input-Process-Output model.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Start with the result	Every problem asks for a useful result: a total, decision, message or control signal. Identify what the result should be first.
Identify specific inputs	Inputs are supplied values such as length and width, a sensor reading or a selected option. Name exact values, rather than saying 'some data'.
Describe the processing	Processing includes calculations, comparisons or decisions. For example: total = unitPrice x quantity, rather than 'the computer works it out'.
Output completes the task	Output could be a displayed cost or warning message. Check that it follows from the processing and answers the stated question.

I Do - One worked example

A program takes a rectangle's length and width and calculates its area. Identify the input, processing and output. [4 marks]

1. The result required is the rectangle's area.

2. Inputs are length and width [1], two values supplied to the algorithm [1].

3. Processing multiplies length x width [1].

4. Output displays the area [1]. A separate result follows the calculation.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A snack shop program calculates the total for five identical snacks. Identify input, processing and output. [4 marks]

2. A temperature monitor displays an ice warning when a reading is below 2°C. Explain input, processing and output. [4 marks]

3. A ticket machine accepts an adult fare and a child fare and displays their total. Give four ordered steps. [4 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the line that prints the pixel count.

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Define processing within an algorithm. [2 marks]

Q3. A streaming service calculates subscription cost from months purchased and a price per month. Describe the IPO stages. [4 marks]

Q4. Explain why identifying IPO accurately helps when designing a school uniform ordering algorithm. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Output is produced after suitable processing has been performed on required inputs. ☐ T ☐ F
2. Every problem asks for a useful result: a total, decision, message or control signal ☐ T ☐ F
3. Inputs are supplied values such as length and width, a sensor reading or a selected option ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Developing a clear, ordered and unambiguous method for solving a problem. ☐

Q2 C - Ciphertext ☐

Q3 It reduces the storage space required for data, using a suitable lossless or lossy method. ☐

Q4 (a) The MAR holds the address being accessed. (b) The MDR holds data or an instruction moving to or from memory. An address is different from the data at it. ☐

Code First - Model

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
print(pixels)
```

Output: 80

We Do - Check the model

1. Inputs: price per snack [1] and quantity 5 [1]. Process: multiply price x 5 [1]. Output: the total price [1]. ☐

2. Temperature from the sensor is input [1]. The program compares it with 2°C [1]. If below the threshold [1], it outputs an ice warning [1]. ☐

3. Input adult fare [1]. Input child fare [1]. Add them [1]. Display the sum [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Operations/calculations performed on inputs [1] to obtain results [1]. ☐

Q3 Inputs are number of months [1] and monthly price [1]. Processing multiplies them [1]. Output displays the cost [1]. ☐

Q4 Inputs could be item types and quantities [1] and unit prices [1]. Processing multiplies unit prices and quantities [1] and totals the order [1]. Output gives total price/confirmation [1]. Precise IPO helps plan/test the algorithm [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: